

Task-Aware Space Warping for Continuous Mixed Reality Collaboration Across Heterogeneous Scenes

Category: Research

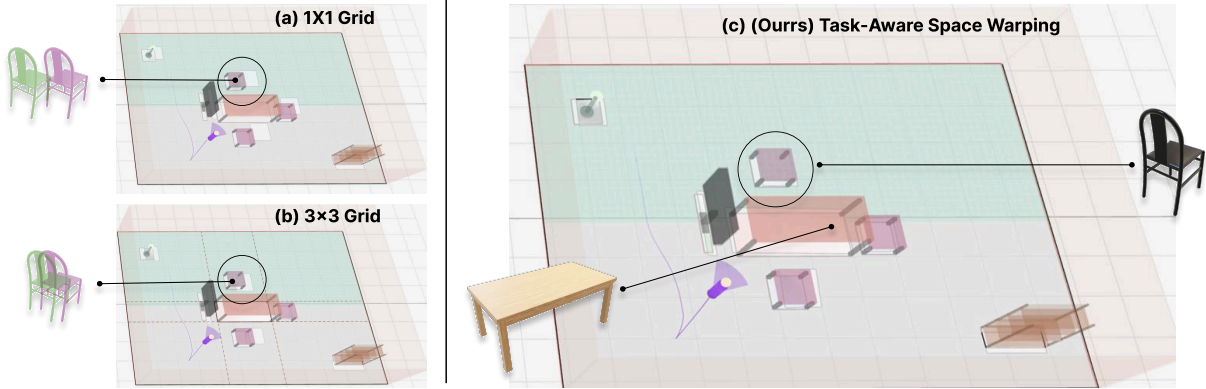


Figure 1: Visual comparison of shared-space generation strategies for a heterogeneous space pair. Left: baseline mappings using (a) 1×1 room-level scaling and (b) task-object-centered 3×3 piecewise mapping. Right: (c) our task-aware space warping, which redistributes distortion according to task relevance to preserve interaction-relevant object structure while maintaining more continuous traversable space between objects.

ABSTRACT

To support continuous and stable mixed reality (MR) collaboration across heterogeneous physical environments, this study proposes task-aware space warping, a pre-mapping spatial deformation technique that distributes distortion and refines the warp field according to the collaboration task. Existing approaches to spatial heterogeneity primarily match shared objects or retarget user positions, often supporting only limited interaction or introducing excessive distortion that can undermine continuous collaboration. To address this, we model an affordance-based constraint field and use it to construct local refinement, global redistribution, and hybrid shared-space warping strategies under different task conditions. Through two evaluations, we compare conventional grid-based mapping and task-aware warping in terms of support for continuous collaboration, and show that even for the same heterogeneous space pair, the desirable shared-space solution may differ depending on the task condition. Overall, this work presents a task-aware pre-mapping framework that preserves the structural usability of task-relevant regions while providing an interactive evaluation simulator for analyzing shared-space generation behavior across methods and conditions.

Index Terms: Virtual Reality, Augmented Reality, Mixed Reality, Spatial Affordance, Heterogeneous Spaces, Space Warping, Remote Collaboration.

1 INTRODUCTION

As mixed reality (MR) collaboration expands beyond co-located demonstrations toward remote collaboration across real physical environments, supporting collaboration between users situated in different places has become increasingly important [31, 33, 51, 52]. In remote MR collaboration, however, users often experience multiple forms of asymmetry, including differences in viewpoint, device setup, embodiment, and surrounding physical environment [29]. Prior work has sought to overcome these asymmetries to support a more coherent shared experience [13, 40]. Among them, heterogeneity in physical space is particularly critical because each user

is embedded in a different room with different geometry, furniture arrangement, and movement constraints. As a result, even when users perform the same collaborative task, the physical support for that task differs substantially across sites, making the construction of a shared collaboration space fundamentally challenging. In practice, collaboration requires more than preserving correspondence at a single shared object: multiple interaction-relevant objects must remain usable, and users must also be able to move between those objects in a stable and continuous way. In this sense, heterogeneous MR collaboration requires a shared-space representation that supports both continuous interaction and continuous movement [27, 38].

To address this challenge, prior work has mainly evolved along two directions. One direction has emphasized object-centered mapping, aiming to preserve correspondence at shared objects or interaction targets [8, 24]. The other has emphasized user-centered retargeting, aiming to maintain navigability or redirect locomotion across mismatched physical spaces [6, 34]. However, these directions often address complementary parts of the same problem rather than the full problem jointly. Object-centered approaches can preserve local interaction targets, but they often do not provide stable locomotion continuity between multiple task-relevant objects. In contrast, user-centered retargeting can preserve locomotion feasibility, but it does not necessarily maintain the stable structure of multiple interaction-relevant objects within a shared spatial configuration [35]. We therefore reformulate shared-space generation in heterogeneous MR collaboration as a problem of jointly balancing object fidelity and locomotion continuity, whose relative importance may vary across collaborative scenarios [7, 14]. Because prior studies have also lacked a consistent analytical basis for comparing these trade-offs [21, 44], our work further frames this problem through an integrated evaluation perspective.

To address this problem, we propose Task-Aware Space Warping. The key idea is to model heterogeneous scenes not merely as geometric layouts but as spatial-affordance-based constraint systems. We represent room boundaries, object footprints, and affordance-aware regions as warping constraints, and then generate

shared space by deforming the space between these constraints. On top of this representation, we define two complementary distortion-allocation strategies. Global Redistribution distributes deformation across the space to promote broader correspondence and room-scale continuity, whereas Local Refinement preserves interaction-relevant regions more directly through local anchor-constrained adjustment. We further combine the two into Hybrid Warping, which enables controllable shared-space outcomes that jointly consider object fidelity and locomotion continuity. These strategies are designed not simply as alternative implementations, but as different ways of allocating distortion between interaction-relevant object regions and the surrounding walkable space.

We evaluate the proposed framework in two stages. First, we compare baseline grid-based methods with the proposed strategies to examine whether task-aware warping is needed beyond conventional mappings. For this comparison, we use complementary metrics that quantify area validity, object preservation, representative path behavior, and fold-related deformation validity, thereby evaluating both interaction support and locomotion continuity within a common analytical framework. The results show that the baselines remain limited in jointly accommodating object-related and locomotion-related requirements, while the proposed methods exhibit clearly different distortion-allocation behaviors; in particular, Hybrid Warping provides a practical compromise between preserving shared object structure and maintaining locomotion-related stability. Second, we examine whether the same heterogeneous space pair should always lead to the same shared-space solution. By varying object-usage structure and interaction priority within the same space pair, we show that desirable shared-space outcomes can change with task condition.

This paper makes three contributions. First, we propose a space warping framework that structures heterogeneous spaces using spatial affordance and geometric constraints, enabling heterogeneous scenes to be represented not merely as mismatched layouts but as deformable constraint fields for shared-space generation. This provides a concrete representation of room boundaries, object footprints, and affordance-aware regions for collaboration across different physical spaces. Second, we introduce a task-aware method that balances object fidelity and locomotion continuity, supporting both continuous interaction and continuous movement within a single shared-space model. Rather than addressing object preservation and locomotion as separate goals, our method formulates them as a jointly adjustable relationship that can vary depending on task context. Third, we present a simulator and evaluation framework that makes these behaviors analyzable and comparable through reproducible measures. Together, these contributions provide a foundation for future MR systems that can generate adaptive, stable, and continuous collaboration spaces from heterogeneous scene models and collaborative context.

2 RELATED WORKS

2.1 Space Mapping Across Heterogeneous Scenes

As one line of research for handling heterogeneous spaces, user-centered retargeting approaches have been proposed to resolve the structural mismatches that arise during collaboration across heterogeneous environments by controlling and adjusting users' visual perception or walking behavior. These approaches have largely re-targeted users to movements or positions that are interpreted as analytically similar across different spaces [15, 41, 50, 49]. In addition, some studies have attempted to address spatial discrepancies by leveraging redirected walking, which induces directional redirection within perceptual thresholds [34, 39, 5, 47, 42, 43]. These approaches have played an important role, particularly in heterogeneous spaces with high dissimilarity, in regulating movement and ensuring users' navigability. However, because these methods focus on the real-time control of user movement, they can

impose cognitive load on users through repeated adjustments and resets during collaboration in complex spaces containing multiple interaction-relevant objects. As a result, they remain limited in directly constructing a stable shared object structure at the spatial level.

As another line of research, pre-mapping approaches have been proposed to generate a shared space by aligning the geometric information of spaces in advance. The core idea of pre-mapping is to identify correspondences between spaces to be used for collaboration and, through them, to construct a collaborative space before the collaboration begins. For example, some studies focused on correspondences between walkable empty areas and proposed empty-space mapping that expands the size of such areas within perceptual thresholds [20, 22, 23]. Methods have also been proposed that pre-map spaces by partitioning them into corresponding affine polygonal patches or grids [24]. In addition, prior work has identified correspondence points between two differently shaped empty spaces and then adjusted users' movement and paths based on that mapping [2]. Although such pre-mapping based on pre-aligned correspondences enables stable interaction during collaboration, it has not sufficiently reflected the complex affordance structure of real scenes because it has been limited to restricted affordances or furniture layouts. Moreover, movement paths and deformation stability in various pre-mapped spaces have not been thoroughly addressed. To address these limitations, our work transforms multiple affordances into constraints and, based on them, proposes space warping that forms a shared field correspondence between heterogeneous spaces.

2.2 Shared Interaction in Mixed Reality Collaboration

In MR remote collaboration, shared interaction can be supported not only through the alignment of room-level spatial structures such as floors and walls [27, 28], but also through the consideration of interior objects [9], which can enhance realism when they remain available for meaningful interaction [10]. By matching interaction-relevant objects, shared interaction can be enabled, and various approaches have been proposed to designate specific objects such as desks, seats, displays, and surfaces as shared targets while maintaining their correspondence [4, 11, 8, 12, 3, 37]. In particular, methods for aligning object edges have also been proposed to support precise and realistic interaction with target collaborative objects [24]. These studies have shown that object preservation is directly connected to collaboration quality, thereby highlighting the importance of structural alignment for shared interaction.

However, prior approaches to shared interaction have primarily focused on single target object correspondence. In real remote collaboration, users do not remain at a single interaction target, but instead use multiple objects and move continuously between them. To support such scenarios, both the alignment of multiple interaction-relevant objects and the locomotion continuity connecting them must be considered together. In response to this need, methods have been proposed to support spatial adjustment and transition in spaces containing multiple furniture objects [7, 14]. There have also been attempts to use scene graphs as a data structure for representing physical environments that enable complex interactions [46], and some studies have explored alignment or space composition based on inter-object relationships represented in scene graphs [30, 25]. However, approaches to shared interaction that jointly consider both multi-object alignment and the continuous flow of movement between objects, in order to support multi-object usage scenarios resembling real collaboration, remain limited. Therefore, this study proposes a space-warping method for continuous collaboration that jointly considers multi-object alignment and inter-object locomotion continuity.

2.3 Evaluation Frameworks for Shared Space

Existing shared-space and remote collaborative MR research has validated system effectiveness through experiential measures such as presence, co-presence, workload, and user experience [45]. Prior studies have examined how furniture arrangement and device configurations in physical environments influence collaborative experience [36], and have also analyzed differences in user experience depending on how collaborative spaces are configured or how furniture objects are utilized [38, 8, 7]. In addition, users' visual representations and movements in collaborative situations have likewise been evaluated through various experiential measures [32, 48, 26]. These user experience-centered evaluations play an important role in understanding actual collaborative experience and in identifying how shared-space systems affect users' cognitive and social experiences. However, such evaluation approaches alone are limited in independently analyzing the geometric precision of the algorithm itself or the structural robustness of the system under diverse physical environmental differences. In particular, because user-centered evaluations involve many controlled variables and require substantial cost to generalize the results, a separate analytical evaluation is needed to precisely compare and analyze the structural properties that arise during shared-space generation.

Accordingly, this study builds on the trajectory of prior work while complementing it by focusing on the engineering-oriented precision analysis of the shared-space generation process. Previous studies have measured the degree of alignment in interactions commonly supported in shared spaces from the perspectives of area, edges, and relational structure [24, 25, 21], and other studies have analyzed the re-adjustment costs required to mitigate dissimilarity [18, 19, 17, 16]. However, because these metrics have been presented according to the goals and scenarios of individual studies, they have remained limited in enabling comparison and interpretation of results under a consistent standard. Moreover, although classifications and discussions of asymmetric interaction and spatial heterogeneity in collaborative spaces have been introduced [1, 44], attempts to organize them into an integrated and systematic analytical framework in the context of collaborative interaction and movement continuity remain limited. To address this gap, this study proposes an analytical evaluation framework that integrates previously fragmented metrics and enables a more consistent analysis of the structural properties and trade-offs that emerge during shared-space generation. Through this framework, we analyze the proposed space warping method under a reproducible engineering validation scheme.

3 METHOD

We propose a task-aware space warping method to simultaneously support continuous movement and object interaction in MR remote collaboration. When multiple corresponding furniture objects exist across heterogeneous physical spaces, we construct semantic and geometric constraints from room and object footprints and generate a continuous shared warp field between the source and target spaces based on these constraints. The core of the proposed method lies in distributing distortion to preserve the usability of interaction-relevant structures while maintaining movement continuity and stability throughout the free space between objects. To this end, we implement task-aware shared-space generation through three complementary strategies (local refinement, global redistribution, and a hybrid strategy combining the two) each of which follows a different distortion-allocation principle.

3.1 Constraint Modeling for Continuous Collaboration

We model continuous collaboration as the coexistence of two spatial requirements: *multi-object alignment* for interaction and *inter-object locomotion continuity* for movement. In remote MR collaboration, users must not only access corresponding objects across

heterogeneous spaces, but also move between those objects without abrupt spatial inconsistency. Accordingly, our goal is not to achieve uniform geometric registration over the entire space, but to construct a shared warp field that preserves task-relevant object structures while maintaining stable free-space traversal.

From the perspective of multi-object alignment, the objective is not merely to place matched objects near one another in the target space, but to preserve their usable interaction structure. In practice, a mapped object region should maintain a consistent footprint, boundary extent, and relative placement so that actions centered on desks, chairs, or displays remain spatially interpretable after warping. From the perspective of locomotion continuity, the key requirement is that deformation should not create abrupt transitions, local collapse, or unstable passage structure in the walkable area between objects. In this sense, locomotion continuity is not satisfied merely by preserving walkable reachability; it additionally requires that local spatial transitions remain gradual enough for users to interpret movement between interaction regions as continuous rather than abruptly reconfigured.

Let the source and target scenes be denoted by S_A and S_B . Each scene is represented on the floor plane as

$$S = (R, O),$$

where R denotes the room boundary and $O = \{o_1, \dots, o_n\}$ denotes the set of objects. Each object o_k is represented by its semantic type, planar position, footprint size, and task relevance level. We represent both rooms and objects by their floor-plane footprints so that the formulation preserves not only object centers but also occupied area and boundary structure relevant to interaction. This footprint-based abstraction is important because collaborative usability often depends on where interaction-supporting surfaces and boundaries remain after warping, not only on whether the object center is approximately aligned.

Based on this abstraction, we define a continuous warp field

$$W : R^2 \rightarrow R^2$$

that maps each source floor-plane point to its shared-space position. The room boundary defines the outer deformation envelope, while each matched object footprint defines an interior constraint region. From these regions, we derive corner and boundary anchors that guide the deformation toward corresponding target structures. Room-level constraints maintain the global spatial frame, whereas object-level constraints preserve interaction-relevant interior structure. As a result, the field is not required to preserve all regions equally; instead, it is guided by a structured set of anchors distributed over both the outer room boundary and the inner object regions.

A key property of the proposed formulation is that object constraints are weighted by task relevance. Not all objects need the same degree of preservation in collaboration. Therefore, each matched object is assigned a task-dependent weight

$$\lambda_k \in \{\lambda^{\text{strong}}, \lambda^{\text{medium}}, \lambda^{\text{weak}}\}, \quad \lambda^{\text{strong}} > \lambda^{\text{medium}} > \lambda^{\text{weak}}.$$

A larger λ_k increases the influence of the corresponding footprint on the warp field, encouraging stronger preservation of its position, extent, and boundary shape. A smaller λ_k allows the surrounding region to absorb more distortion in favor of room-scale continuity. In practice, this weighting affects both how strongly a region is pulled toward its target footprint and how strongly its boundary structure resists local deformation, thereby controlling the local distribution of distortion around task-relevant objects.

Under this formulation, continuous collaboration is achieved not by perfectly preserving all geometry, but by distributing distortion according to collaboration relevance. The resulting constraint

model therefore encodes where the shared space should remain structurally faithful for object interaction and where deformation can be allocated more flexibly to support locomotion continuity. Based on this model, we construct the shared warp field using three complementary strategies that differ in how they allocate distortion across the room.

3.2 Space Warping Strategies

3.2.1 Local Refinement

The local refinement strategy constructs the warp field over a regular grid defined on the source floor plane. Let

$$W_{ij} = (X_{ij}, Z_{ij})$$

denote the mapped position of grid node (i, j) . The continuous field is obtained by bilinear interpolation over these nodes. The field is initialized by a coarse room-to-room mapping and then iteratively refined using room and object anchors.

For each non-fixed node, the update balances local smoothness and anchor fidelity. Let

$$\bar{W}_{ij} = \frac{1}{4} (W_{i-1,j} + W_{i+1,j} + W_{i,j-1} + W_{i,j+1})$$

be the local smoothness target, and let $\hat{W}_{ij}$ be the soft-anchor target with weight λ_{ij} . The update is defined as

$$W_{ij}^* = \arg \min_w (\mu \|w - \bar{W}_{ij}\|^2 + \lambda_{ij} \|w - \hat{W}_{ij}\|^2),$$

which yields the weighted average

$$W_{ij}^* = \frac{\mu \bar{W}_{ij} + \lambda_{ij} \hat{W}_{ij}}{\mu + \lambda_{ij}}.$$

Thus, nodes near task-critical anchors are pulled more strongly toward object correspondence, whereas less constrained regions absorb distortion more flexibly through smooth interpolation.

To preserve object usability beyond pointwise matching, local refinement additionally regularizes sampled room and object boundaries so that mapped boundary samples remain coherent and approximately aligned with their corresponding target edges. This is important because many collaborative interactions depend not only on object location but also on the preservation of usable boundary extent, such as where a desk edge, chair footprint, or display-facing side is expected to remain after warping. Accordingly, local refinement emphasizes object-level fidelity by preserving both anchor correspondence and local boundary structure.

To avoid invalid deformation, the method evaluates the signed area of each grid cell and suppresses updates that increase local fold-over. As a result, local refinement produces a field that closely preserves interaction-relevant anchors while remaining continuous and locally valid over the room.

3.2.2 Global Redistribution

Whereas local refinement concentrates deformation around anchor neighborhoods, global redistribution allocates distortion more broadly across the room. Its purpose is to preserve locomotion continuity by reducing excessive local compression and spreading deformation over the full walkable region.

To do this, we construct ordered breakpoint sets along the horizontal and vertical axes using room boundaries and matched object footprint extents. Let b^{global} denote the room-scale destination of a breakpoint and b^{target} denote its corresponding object-aware target. The final breakpoint destination is defined as

$$b^{\text{dst}} = (1 - \beta) b^{\text{global}} + \beta b^{\text{target}},$$

where β controls the extent to which object correspondence influences the global redistribution.

Using these ordered breakpoint pairs, we construct monotone axis-wise interpolation functions

$$x' = f_x(x), \quad z' = f_z(z),$$

and define the warp as

$$W(x, z) = (f_x(x), f_z(z)).$$

This separable formulation yields a globally ordered deformation that spreads distortion across the room while maintaining monotonicity along each axis. Because deformation is distributed in an ordered axis-wise manner rather than concentrated around isolated anchor neighborhoods, the resulting field better preserves broad spatial progression across the room and is therefore more suitable for locomotion-oriented scenarios.

Room corners and object corners are then reimposed to stabilize the field, and weak structural attraction is added around object boundaries. Consequently, global redistribution prioritizes room-scale continuity and is particularly suitable when stable inter-object movement is more important than strong local fitting around every task-relevant object.

3.2.3 Hybrid Space Warping

The hybrid strategy combines the complementary strengths of the previous two strategies. A purely local strategy can preserve task-relevant anchors well but may concentrate distortion too sharply, whereas a purely global strategy can preserve broad continuity but may underfit critical object structures. To balance these requirements, the hybrid method adopts a coarse-to-fine formulation.

First, a global redistribution field is constructed as an initial coarse warp,

$$W^{(0)} = W_{\text{global}}.$$

Then, local refinement is applied using this field as initialization,

$$W^* = \text{Refine}(W^{(0)}).$$

In this way, the global stage first distributes distortion at the room scale, and the local stage then recovers object-level fidelity around task-relevant anchors.

An important property of the hybrid formulation is that it can generate different shared-space solutions by adjusting the relative emphasis between the global and local stages. By tuning parameters that control breakpoint blending, anchor attraction, smoothness, and boundary regularization, the method can realize navigation-oriented, object-focused, or balanced behaviors. Navigation-oriented settings place greater emphasis on room-scale redistribution and gradual spatial transition, whereas object-focused settings increase the influence of task-relevant anchors and boundary preservation. Balanced settings seek an intermediate trade-off between the two.

Accordingly, the hybrid formulation is not introduced merely as a compromise, but as a mechanism for exposing multiple valid shared-space solutions under different collaborative priorities. The same pair of heterogeneous spaces does not necessarily lead to a single desirable shared-space result; instead, different collaborative tasks may call for different distortion-allocation outcomes. The hybrid strategy makes this design space explicit within a unified task-aware warp-field framework.

4 EVALUATION

We evaluated the proposed method with two goals. First, we examined whether task-aware warping strategies provide more suitable shared-space behavior than simple grid-based mappings when both

object interaction and movement continuity must be considered together. Second, we examined whether different task-aware warping configurations produce different shared-space outcomes even for the same heterogeneous space pair. To this end, we conducted two studies using a representative heterogeneous space pair and compared baseline mappings with the proposed local refinement, global redistribution, and hybrid space warping methods.

4.1 Evaluation Framework

To evaluate shared-space quality, we used geometric metrics that reflect object preservation, walkable-space validity, path continuity, and deformation stability. Rather than relying on a single score, we measured several complementary properties of the generated shared space, since continuous collaboration requires both usable interaction structure and stable locomotion-supporting free space.

Area and object metrics. We first measured whether the warped shared space preserved valid semantic occupancy across the target floor plane. For this purpose, target-space samples were uniformly drawn over the floor plane, excluding points outside the room boundary. Each sample was then classified according to whether it was covered by a semantically matched warped source region and whether it belonged to a folded region.

Based on this procedure, we computed *Valid Area Ratio*, defined as the proportion of target-space samples that were covered by a semantically consistent warped source region and did not fall in a folded area. This metric indicates how much of the target shared space remains valid for collaboration after warping. We also computed *Dead Area Ratio*, defined as the proportion of target samples that became unusable because of fold-over, semantic mismatch, or non-overlap. This metric reflects how much area is lost or invalidated by the deformation.

To evaluate object preservation more directly, we measured both object-interior and object-boundary consistency. *Object Plane Mapping Rate* was computed by densely sampling each source object footprint and checking whether the warped samples landed inside the corresponding target object footprint without entering folded regions. This metric indicates how well object area is preserved under warping. *Object Edge Mapping Rate* was computed by sampling points along source object boundaries and checking whether the warped samples remained close to the corresponding target object boundary within a fixed distance tolerance. This metric reflects how well object boundary structure is preserved, which is important for interaction-relevant surfaces and edges.

We additionally measured *Walkable Area Mapping Rate*, defined as the proportion of source walkable samples that remained walkable after warping in the target space. This metric captures whether free space supporting locomotion is preserved as navigable shared space.

Path and smoothness metrics. To evaluate movement continuity, we defined a small set of representative paths on the source floor plane, consisting of horizontal, vertical, and diagonal traversals across the room. These paths were not derived from user trajectories, but were used as controlled geometric probes to compare how each mapping transforms room-scale movement structure.

For these representative paths, we computed *Path Mapping Rate*, defined as the proportion of warped path samples that remained walkable and non-folded in the target space. This metric indicates whether a mapped path remains traversable. We also computed *Path Length Distortion*, defined as

$$\left| \frac{L_{\text{mapped}}}{L_{\text{source}}} - 1 \right|,$$

where L_{source} and L_{mapped} denote the source and warped path lengths, respectively. This metric captures how much the overall path length changes under warping.

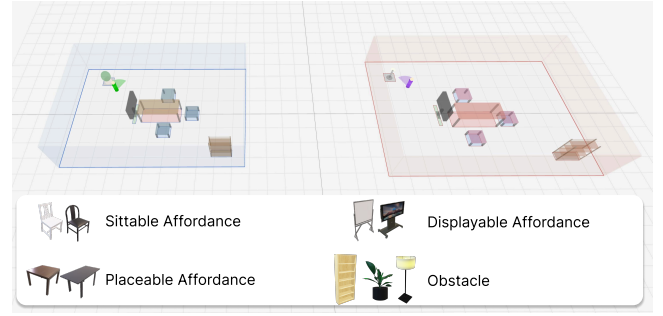


Figure 2: Representative heterogeneous space pair used for evaluation, illustrating differences in room geometry, furniture layout, and interaction-relevant object arrangement.

To further quantify path-shape preservation, we computed *Path Shape Deviation*. For each representative path, the warped path and a room-scaled reference path were resampled with the same number of points, and the mean pointwise distance between the two was measured. This metric reflects how strongly the path geometry deviates from a globally scaled reference trajectory.

Finally, we measured *Smoothness Cost* to quantify how abruptly local deformation changes across the walkable region. For sampled walkable locations, we first estimated a local deformation cost based on stretch and anisotropy, and then computed the mean absolute difference of this cost between neighboring samples. A lower smoothness cost indicates that local distortion changes more gradually across space, which is desirable for movement continuity.

Fold and validity metrics. To evaluate deformation validity, we measured fold-related metrics on the warped grid. A grid cell was regarded as folded or degenerate when its signed area became non-positive or nearly zero after deformation. Based on this criterion, we computed *Fold Count*, which counts the number of invalid warped cells, and *Fold Area*, which measures the spatial extent of such invalid regions. These metrics indicate whether the generated warp remains geometrically usable as a continuous shared-space field.

4.2 Experimental Setup

We evaluated all methods on a representative heterogeneous space pair containing multiple corresponding furniture objects with different interaction relevance. The source and target spaces differed in room dimensions, object placement, and footprint extents, while preserving object correspondence across the pair. We used this fixed space pair throughout both studies so that observed differences reflect the properties of the compared warping strategies rather than differences in scene composition.

We compared two baseline mappings and three proposed methods. The first baseline, *1×1 room-level scaling*, uniformly scales the source room to the target room as a single global region. The second baseline, *task-object-centered 3×3 piecewise mapping*, partitions the room around the primary task object and maps each region separately in a coarse piecewise manner. The proposed methods were *local refinement*, *global redistribution*, and *hybrid space warping*. In Study 1, we compared the two baselines against local refinement, global redistribution, and hybrid space warping under a desk-only strong condition. In Study 2, we fixed the same space pair and compared multiple hybrid configurations under different task-emphasis settings, including desk-only strong, all-strong, all-weak, and desk-strong with chair/display-medium conditions.

All methods were implemented in a web-based simulator with a Python backend and a JavaScript frontend. Warp-field computation was implemented in Python using FastAPI, and interactive visu-

Table 1: Study 1 results comparing baseline mappings and proposed warping strategies under the desk-only strong condition. Higher is better for Valid, ObjA, ObjE, Walk, and Path. Lower is better for pDist, Smooth, and Folds. Best values in each column are shown in bold.

Method	Valid	ObjA	ObjE	Walk	Path	pDist	Smooth	Folds
1×1 Grid	0.963	0.357	0.042	0.958	0.790	0.152	0.000	0
3×3 Grid	0.981	0.635	0.342	0.977	0.729	0.153	0.004	0
Local Refinement	0.989	0.701	0.398	0.981	0.735	0.169	0.032	0
Global Redistribution	0.999	0.909	0.893	0.989	0.741	0.167	0.197	2
Hybrid Tuning	0.996	0.752	0.556	0.986	0.741	0.170	0.034	0

alization and metric comparison were implemented in JavaScript using Three.js. All development and evaluation were conducted on a desktop PC with an Intel Core i9-13900KF CPU, NVIDIA RTX 4090 GPU, 32 GB RAM, and 4 TB storage.

Because our goal was to compare geometric behaviors of different shared-space generation strategies rather than to make population-level statistical claims, we report the resulting metric values descriptively. Accordingly, the results below focus on relative tendencies and observed trade-offs among methods, without statistical significance testing.

4.3 Study 1: Baseline Comparison

4.3.1 Compared Methods

In Study 1, we compared two baseline mappings with the three proposed methods under a desk-only strong condition. The baselines were 1×1 room-level scaling and task-object-centered 3×3 piecewise mapping. The proposed methods were local refinement, global redistribution, and hybrid space warping. This comparison was intended to examine how different distortion-allocation strategies affect object preservation, walkable-space validity, and path-related continuity within the same heterogeneous space pair.

4.3.2 Results

Table 1 shows the results of Study 1. Overall, all proposed methods produced high valid-area ratios and low dead-area ratios, while differing in how strongly they favored object preservation, locomotion-related continuity, and deformation validity.

Among all compared methods, global redistribution showed the highest Valid Area Ratio (0.999), the lowest Dead Area Ratio (0.001), the highest Object Plane Mapping Rate (0.909), the highest Object Edge Mapping Rate (0.893), and the highest Walkable Area Mapping Rate (0.989). It also produced the highest Path Mapping Rate (0.741), together with hybrid space warping. However, global redistribution also showed the largest Path Shape Deviation (0.457), the highest Smoothness Cost (0.197), and a nonzero Fold Count (2).

Local refinement improved object-related metrics over both baselines, yielding an Object Plane Mapping Rate of 0.701 and an Object Edge Mapping Rate of 0.398, while maintaining zero folds. Compared with 1×1 room-level scaling and 3×3 piecewise mapping, it also showed higher Valid Area Ratio (0.989) and Walkable Area Mapping Rate (0.981). At the same time, its Path Mapping Rate (0.735) remained close to that of the baselines.

Hybrid space warping showed an intermediate profile between global redistribution and local refinement. It yielded a high Valid Area Ratio (0.996), a low Dead Area Ratio (0.004), zero folds, and the joint-highest Path Mapping Rate (0.741). Its object-related metrics (0.752 for Object Plane Mapping Rate and 0.556 for Object Edge Mapping Rate) were higher than those of local refinement but lower than those of global redistribution. In addition, its Smoothness Cost (0.034) remained much lower than that of global redistribution and close to that of local refinement.

Compared with the proposed methods, the two baselines showed lower object-preservation performance. In particular, 1×1 room-level scaling showed the lowest Object Plane Mapping Rate (0.357)

and Object Edge Mapping Rate (0.042), while 3×3 piecewise mapping improved these values to 0.635 and 0.342, respectively. Both baselines produced zero folds and low smoothness cost, but their object-related preservation remained lower than that of the proposed methods.

Overall, the Study 1 results show that the compared methods produce different performance profiles even under the same space pair and task condition. Global redistribution favored broader area and object coverage but introduced some fold-related invalidity and higher deformation variation, local refinement preserved object-related structure with stable validity, and hybrid space warping yielded a more balanced result across these metrics.

4.4 Study 2: Task-Aware Hybrid Space Warping

4.4.1 Compared Conditions

In Study 2, we focused on the hybrid space warping method and compared three hybrid presets—navigation, balanced, and object-focused—under four task-emphasis conditions: desk-only strong, all-strong, all-weak, and desk-strong with chair/display-medium. The purpose of this study was to examine whether different task-aware settings produce different shared-space outcomes even when the underlying space pair remains unchanged.

4.4.2 Results

Across all four task-emphasis conditions, the three hybrid presets produced closely related but distinguishable metric profiles. In general, the object-focused preset tended to produce the highest object-related and area-related scores, the navigation preset tended to produce the lowest path-length distortion and path-shape deviation, and the balanced preset remained between the two.

Under the desk-only strong condition, the object-focused preset showed the highest Valid Area Ratio (0.998), the lowest Dead Area Ratio (0.002), the highest Object Plane Mapping Rate (0.763), the highest Object Edge Mapping Rate (0.599), and the highest Walkable Area Mapping Rate (0.989). The navigation preset showed the lowest Path Length Distortion (0.167) and the lowest Path Shape Deviation (0.273). The balanced preset remained between these two in most metrics.

A similar pattern appeared under the all-strong condition. The object-focused preset showed the highest Object Plane Mapping Rate (0.821) and the lowest Path Shape Deviation (0.300), while the balanced preset showed the highest Object Edge Mapping Rate (0.785), the highest Walkable Area Mapping Rate (0.990), and the highest Path Mapping Rate (0.743). The navigation preset showed the lowest Path Length Distortion (0.174) but also lower object-related scores than the other two presets.

Under the all-weak condition, the object-focused preset again showed the highest Valid Area Ratio (0.996), the lowest Dead Area Ratio (0.004), the highest Object Plane Mapping Rate (0.760), the highest Object Edge Mapping Rate (0.588), and the highest Walkable Area Mapping Rate (0.984). The navigation preset showed the lowest Path Length Distortion (0.166) and the lowest Path Shape Deviation (0.272). The balanced preset remained intermediate, with values close to both neighboring presets.

Under the desk-strong with chair/display-medium condition, the object-focused preset again showed the highest Valid Area Ratio (0.998), the lowest Dead Area Ratio (0.002), the highest Object Plane Mapping Rate (0.781), the highest Object Edge Mapping Rate (0.613), the highest Walkable Area Mapping Rate (0.989), and the highest Path Mapping Rate (0.743). The navigation preset showed the lowest Path Length Distortion (0.168) and the lowest Path Shape Deviation (0.273). The balanced preset again lay between the other two presets on most metrics.

Notably, all hybrid presets produced zero folds in all four task conditions. Thus, the main differences among presets appeared not

Table 2: Study 2 results comparing hybrid presets under different task-emphasis conditions. Higher is better for Valid, ObjA, ObjE, Walk, and Path. Lower is better for pDist, Smooth, and Folds. Best values within each condition are shown in bold.

Condition	Hybrid Preset	Valid	ObjA	ObjE	Walk	Path	pDist	Smooth	Folds
3*Desk only strong	Navigation	0.992	0.731	0.463	0.984	0.738	0.167	0.034	0
	Balanced	0.996	0.752	0.556	0.986	0.741	0.170	0.034	0
	Object-focused	0.998	0.763	0.599	0.989	0.741	0.171	0.037	0
3*All strong	Navigation	0.998	0.755	0.709	0.989	0.741	0.174	0.088	0
	Balanced	0.997	0.813	0.785	0.990	0.743	0.177	0.049	0
	Object-focused	0.996	0.821	0.780	0.989	0.738	0.181	0.042	0
3*All weak	Navigation	0.993	0.736	0.466	0.983	0.735	0.166	0.032	0
	Balanced	0.993	0.749	0.534	0.983	0.741	0.169	0.032	0
	Object-focused	0.996	0.760	0.588	0.984	0.741	0.171	0.034	0
3*Desk strong / Chair+Display medium / Others weak	Navigation	0.992	0.733	0.480	0.985	0.738	0.168	0.034	0
	Balanced	0.996	0.752	0.559	0.986	0.741	0.171	0.034	0
	Object-focused	0.998	0.781	0.613	0.989	0.743	0.172	0.039	0

in deformation validity, but in how area preservation, object preservation, and path-related distortion were balanced. Overall, the Study 2 results show that even within the same hybrid framework and the same space pair, different task-aware settings lead to different shared-space outcomes across object-related and locomotion-related metrics.

5 DISCUSSION

5.1 Metric-Profile Analysis of Shared-Space Warping

The Study 1 results suggest that the proposed space warping family better accommodates both object-related and locomotion-related requirements than simple grid-based baselines. Rather than excelling on only a single metric dimension, the proposed methods produced more structured trade-offs across object preservation, walkable-space validity, path-related behavior, and deformation stability. In contrast, the two baselines showed more limited metric profiles. In particular, 1×1 room-level scaling showed very low object-related preservation, and although the task-object-centered 3×3 piecewise mapping improved object metrics, it still remained below the proposed methods in object-plane and object-edge preservation. This pattern suggests that conventional grid-based mappings are limited in how well they jointly accommodate interaction-relevant object structure and locomotion-supporting continuity.

The difference between local refinement and global redistribution is better understood not as a simple parameter change, but as a difference in distortion-allocation principle. Local refinement emphasizes anchor-constrained smooth adjustment and therefore showed a more continuity-stable profile, with zero folds and relatively low smoothness cost. By contrast, global redistribution more strongly favored correspondence-oriented behavior, producing the highest object-related metrics and the highest valid-area ratio in Study 1. At the same time, it also incurred higher smoothness cost and nonzero folds, indicating that stronger room-scale redistribution can improve correspondence while introducing additional deformation-side cost. In this sense, the two strategies reveal different ways of distributing distortion across shared space: local refinement preserves local stability around anchors, whereas global redistribution more aggressively reallocates distortion to improve room- and object-level coverage.

Hybrid space warping should likewise not be interpreted as a simple average between the two. Because it first distributes distortion globally and then refines the field locally, it forms a controllable compromise between correspondence-oriented and continuity-oriented behavior. In Study 1, hybrid space warping preserved substantially higher object fidelity than the baselines while avoiding the folds introduced by global redistribution and main-

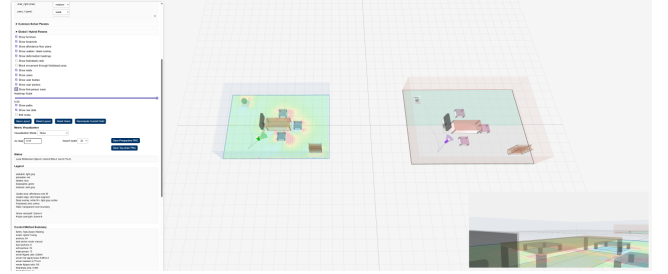


Figure 3: Screenshot of interactive simulator for task-aware space warping, showing scene configuration, warping strategy selection, and metric-based visualization of shared-space generation.

taining a smoothness cost close to that of local refinement. This intermediate profile is especially important in light of Study 2.

Study 2 shows that even within the same heterogeneous space pair, different task-emphasis settings lead to different desirable shared-space outcomes. Under desk-only strong, all-strong, all-weak, and hierarchical priority conditions, the three hybrid presets produced systematically different metric profiles. Across these conditions, the object-focused preset tended to favor higher object-related and area-related preservation, whereas the navigation-oriented preset tended to yield lower path-length distortion and lower path-shape deviation. The balanced preset typically remained between these two behaviors. These results suggest that the desirable shared-space result is not determined only by the scene pair itself, but also by how collaborative importance is distributed across objects.

Taken together, these findings suggest that shared-space generation is better understood as a task-aware solution space than as a fixed mapping problem. Even for the same pair of heterogeneous spaces, different collaboration priorities may call for different distortion-allocation outcomes. The proposed formulation makes this design space explicit by allowing object preservation and locomotion continuity to be rebalanced through task-aware constraints and hybrid tuning.

5.2 Interactive Evaluation Framework

The proposed method and evaluation are implicated as a simulator that supports interactive analysis of task-aware shared-space generation by combining scene editing, task-priority assignment, warp computation, and metric visualization within a single environment (Figure 3). Through the interface, users can change scene layout, object task levels, warping strategy, and hybrid preset, and

then inspect the resulting warp both visually and quantitatively. In addition to top-down and perspective visualization, the simulator provides overlays for valid/dead regions, fold cells, deformation heatmaps, representative paths, and smoothness gradients, together with object-, area-, path-, and fold-related metrics. Accordingly, the simulator functions not only as a visualization tool but also as an analysis environment for examining the structural behavior of different warping strategies under controlled conditions.

In this sense, the current framework can serve as a reusable basis for future studies on task-aware MR shared-space generation. Because the same scene pair can be evaluated under multiple task-emphasis settings and solver configurations, it enables systematic comparison of how different strategies produce different metric profiles and geometric behaviors. Beyond the present study, the framework could be extended toward parameter tuning, benchmark construction, recommendation-based setting selection, and adaptive shared-space generation experiments in which different warp solutions are explored under changing collaboration priorities.

5.3 Limitations

While this study evaluated the proposed method on a representative heterogeneous space pair, broader generalization across more diverse scene pairs remains to be verified. The present experiments were designed to reveal the structural behavior of task-aware warping strategies under controlled conditions rather than to exhaustively characterize performance across all possible room geometries and object distributions. In addition, the current task conditions were modeled mainly through object-usage structure and interaction-priority assignment. Although this formulation was sufficient to expose meaningful differences in distortion-allocation behavior, it does not yet capture richer collaborative semantics, dynamic task transitions, or overlapping interaction demands that may arise in real collaboration.

Another limitation is that the current evaluation remains primarily analytical and simulator-based. The proposed metrics characterize geometric behavior, object preservation, walkable validity, and deformation stability, but they do not directly establish how these properties map onto actual collaborative experience. Moreover, while the simulator is suitable for comparing shared-space generation behavior under controlled conditions, it does not yet incorporate all real-world system factors, such as sensing noise, incomplete spatial models, real-time adaptation constraints, or richer social interaction dynamics. Additional validation under more realistic deployment conditions, as well as stronger links to user-centered MR collaboration studies, will therefore be necessary to assess system-level applicability.

6 CONCLUSION AND FUTURE WORK

This study proposed task-aware space warping for MR remote collaboration across heterogeneous physical spaces. By modeling room and object footprints as spatial constraints with different levels of task relevance, we constructed shared warp fields using three complementary strategies: local refinement, global redistribution, and hybrid space warping. Through analytical evaluation on a representative heterogeneous space pair, we showed that these strategies produce different metric profiles across object preservation, walkable space validity, path continuity, and deformation stability. The results suggest that shared-space generation is better understood not as a single fixed mapping problem, but as a space of possible solutions whose desirable outcome depends on collaboration priorities. We also presented an interactive evaluation simulator that supports controlled comparison of scene configurations, task conditions, warping strategies, and resulting metric profiles within a unified framework.

Future work can extend the present framework in several directions. First, the proposed method should be evaluated across more

diverse space pairs and object configurations to better understand how its behavior generalizes. Second, the current task model can be expanded toward richer collaborative semantics, dynamic task transitions, and changing object priorities over time. Third, stronger connections are needed between the analytical metrics used in this study and actual collaborative experience through user studies in MR settings. Finally, the current framework could be extended toward adaptive shared-space generation in which the system selects or adjusts a suitable warp strategy according to scene structure, task condition, and collaboration goals, either from explicit user input or from an automated recommendation process.

DISCLOSURE ON GEN AI USAGE

In accordance with IEEE guidelines, the authors used OpenAI ChatGPT for developing methods and an evaluation framework, and for visualizing data and linguistic editing. All scientific content, experimental design, and data analysis remain the sole responsibility of the authors.

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